ETHAN SETH S. GRATUITO
BS Chemical Engineering 2A

CHEM 13 - ORGANIC CHEMISTRY

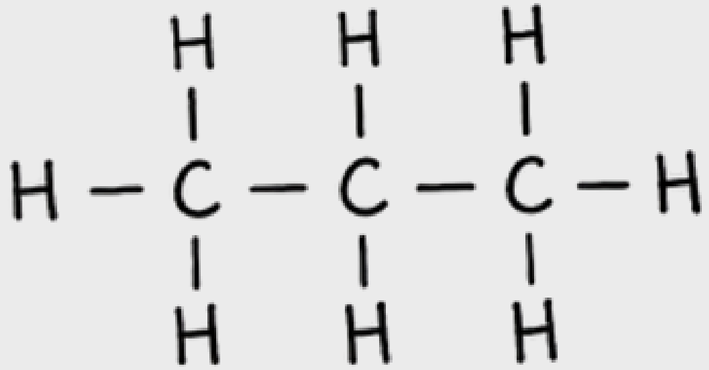
# one molecule, four representations

# Card 1 – Propane ($C_3H_8$)

## 1. Molecular Formula

$$C_3H_8$$

## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3CH_2CH_3$$

## 4. Bond-Line Formula (Ketcher)



## Verification Strip

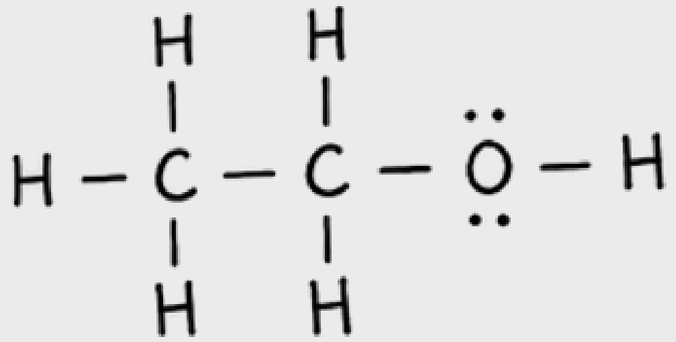
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 2 – Ethanol ($C_2H_6O$)

## 1. Molecular Formula

$$C_2H_6O$$

## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3CH_2OH$$

## 4. Bond-Line Formula (Ketcher)




## Verification Strip

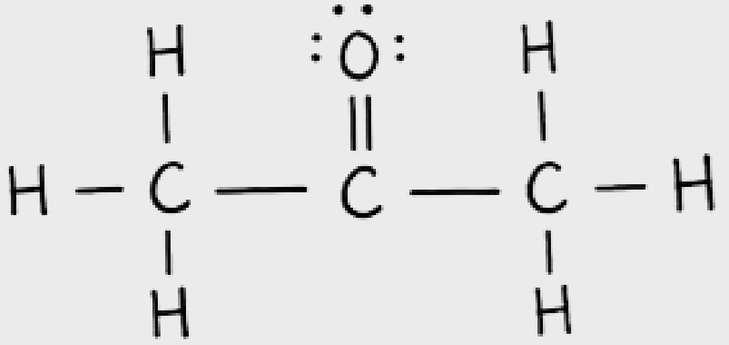
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 3 – Acetone ($C_3H_6O$)

## 1. Molecular Formula

$$C_3H_6O$$

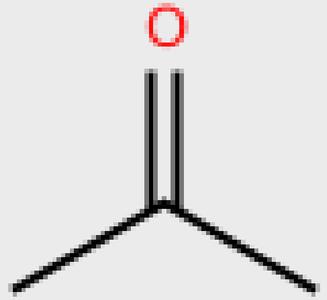
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3COCH_3$$

## 4. Bond-Line Formula (Ketcher)



## Verification Strip

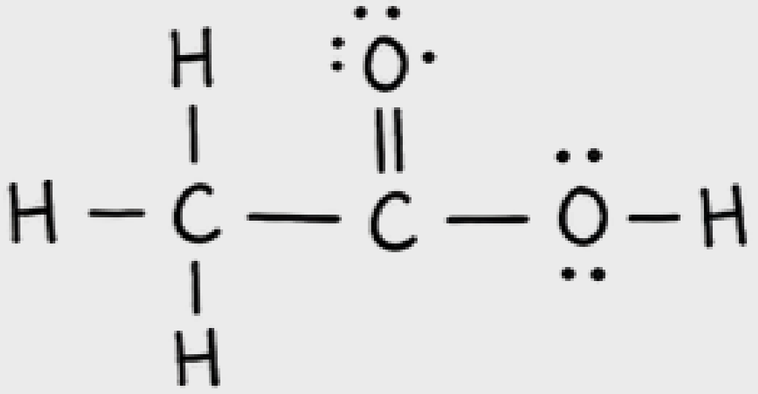
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 4 – Acetic Acid ($C_2H_4O_2$)

## 1. Molecular Formula

$$C_2H_4O_2$$

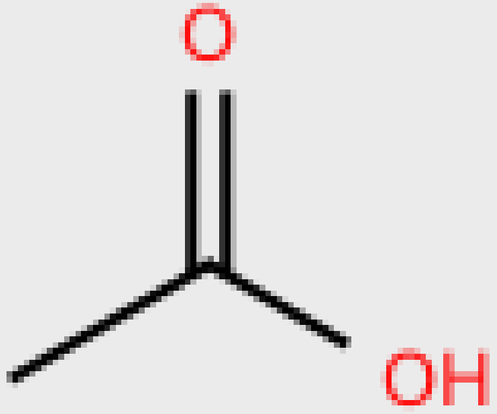
## 2. Lewis Structure




## 3. Condensed Structural Formula

$$CH_3COOH$$

## 4. Bond-Line Formula (Ketcher)




## Verification Strip

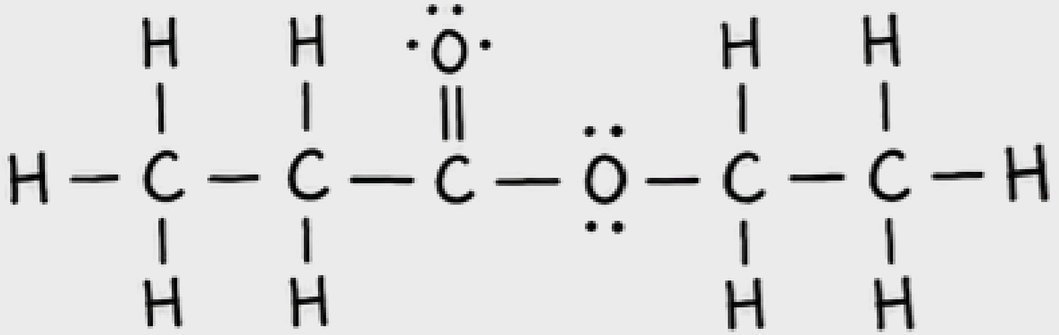
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 5 – Ethyl Acetate ($C_4H_8O_2$)

## 1. Molecular Formula

$$C_4H_8O_2$$

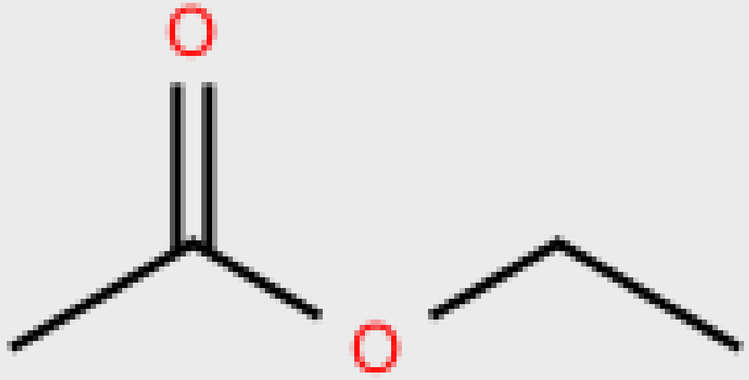
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3COOCH_2CH_3$$

## 4. Bond-Line Formula (Ketcher)



## Verification Strip

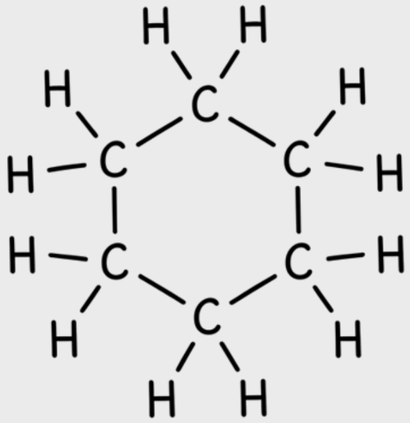
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 6 – Cyclohexane ($C_6H_{12}$)

## 1. Molecular Formula

$$C_6H_{12}$$

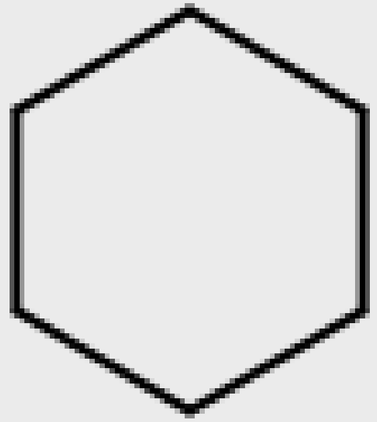
## 2. Lewis Structure




## 3. Condensed Structural Formula

$$(CH_2)_6$$

## 4. Bond-Line Formula (Ketcher)



## Verification Strip

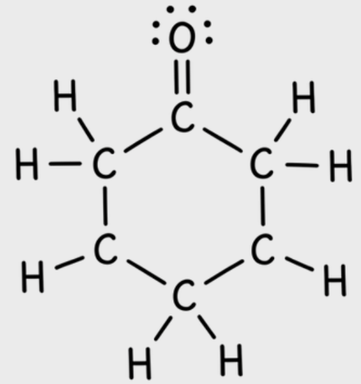
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 7 - Cyclohexanone ($C_6H_{10}O$)

## 1. Molecular Formula

$$C_6H_{10}O$$

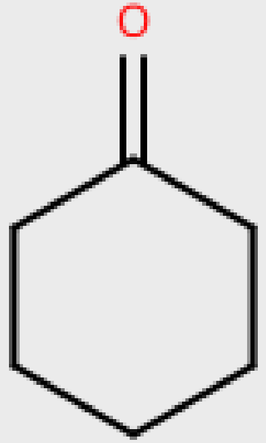
## 2. Lewis Structure




## 3. Condensed Structural Formula

$$(CH_2)_5CO$$

## 4. Bond-Line Formula (Ketcher)




## Verification Strip

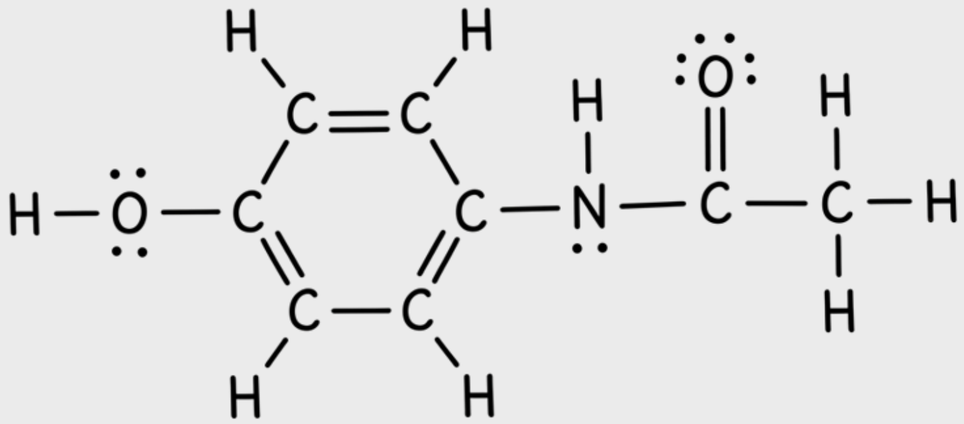
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 8 – Acetaminophen ($C_8H_9NO_2$)

## 1. Molecular Formula

$$C_8H_9NO_2$$

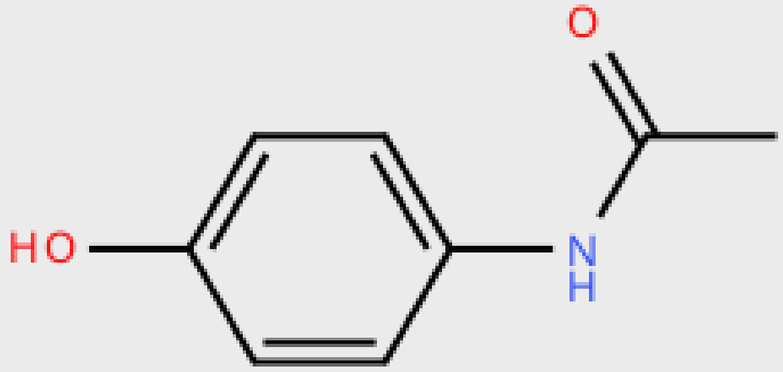
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$HO{-}C_6H_4{-}NHCOCH_3$$

## 4. Bond-Line Formula (Ketcher)



## Verification Strip

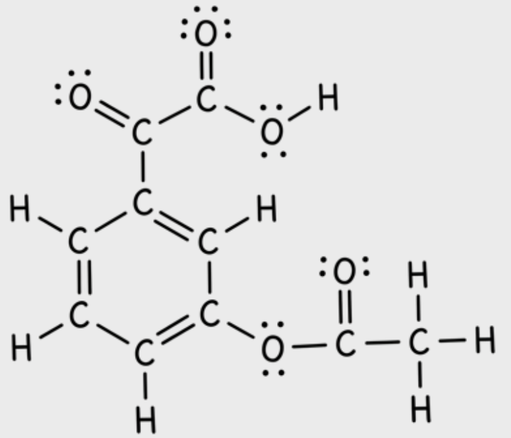
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 9 – Aspirin ($C_9H_8O_4$)

## 1. Molecular Formula

$$C_9H_8O_4$$

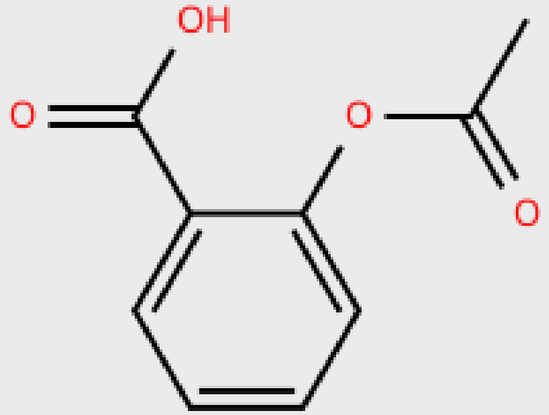
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3CO_2\text{–}C_6H_4\text{–}COOH$$

## 4. Bond-Line Formula (Ketcher)




## Verification Strip

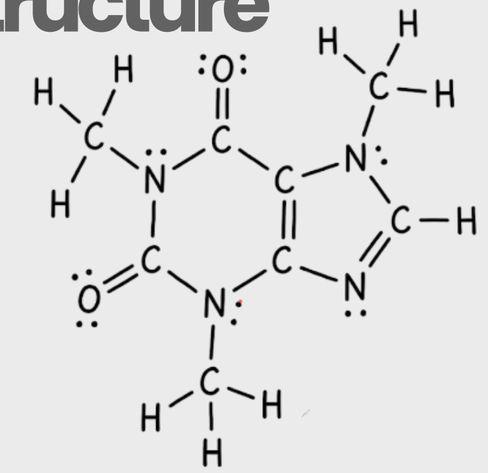
- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

# Card 10 – Caffeine ($C_8H_{10}N_4O_2$)

## 1. Molecular Formula

$$C_8H_{10}N_4O_2$$

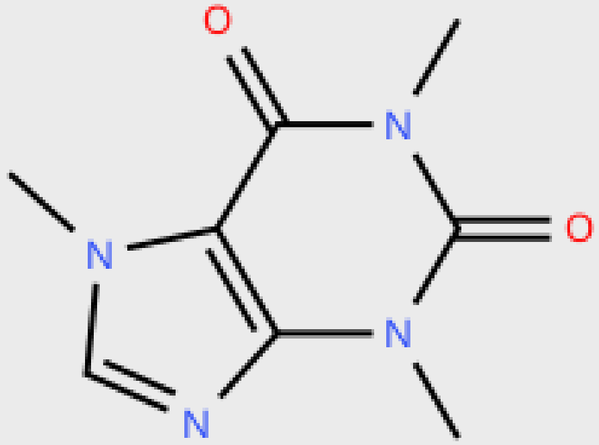
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$C_8H_{10}N_4O_2$$

## 4. Bond-Line Formula (Ketcher)




## Verification Strip

- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed